DETSI Moreton Bay Report

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1 Executive Summary

Seagrass meadows are a critical component of the Moreton Bay ecosystem, providing habitat structure for a wide array of important marine species like dugongs and green turtles. This report presents a spatially explicit statistical analysis of total seagrass cover in Moreton Bay across three financial years (2020–2021, 2022–2023, and 2023–2024), spanning a major flood event that caused widespread seagrass loss. Using water quality data from the eReefs monitoring platform alongside field observations and geographic data, we address three management-relevant research questions: What water quality conditions trigger concern for seagrass cover, whether and where cover has recovered since the flood, and what site-level characteristics make seagrass meadows more or less resilient to disturbance.

Our results indicate that ambient turbidity is a strong water quality driver of total seagrass cover, with a one-unit increase in 90-day mean turbidity associated with a 7.18 percentage point decrease in cover — a relationship that holds across the bay but is more arguably consequential in intertidal and mixed-sediment environments where cover is naturally highest. Flood-driven turbidity also negatively affected cover, though its impact diminished with time since the flood. Bay-wide, seagrass cover showed measurable recovery between 2022–2023 and 2023–2024, but this recovery was spatially uneven: EHMP subzones including Deception Bay and South-Eastern Moreton Bay returned to pre-flood levels, while Southern Bay in particular continued to lag; Conseravtion zones including Conservation Park Zone and Marine National Park Zone returned to nearly pre-flood levels, while General Use Zone had the least amount of negative impact from the flood. Intertidal sites recovered less effectively than deep subtidal or shallow subtidal sites. At the site level, seagrass dominated by CS or ZMHU species, having high species richness, or in deep waters tended to be most resilient. Together, these findings can help inform areas where seagrass conservation investment is likely to be most effective.

2 Background

Accurately characterizing drivers and resilience of seagrass cover is essential to improving the ecological health of the Moreton Bay ecosystem. Over the past few decades, various scientific efforts have informed our current understanding of trends, behaviors, and flood responses of seagrass cover in Moreton Bay [Abal and Dennison, 1996, Udy and Dennison, 1997, Samper-Villarreal et al., 2016, Ovsyanikova et al., 2026] as well as Australia more widely [Kilminster et al., 2015, Gilby et al., 2018, O'Brien et al., 2018]. Remote sensing has offered new capabilities for seagrass cover mapping [Roelfsema and Phinn, 2009], while innovative statistical approaches have offered novel methods for studying resilience at site-specific scales [Gilby et al., 2023]. Our work contributes to this body of literature by applying spatially explicit statistical models [Cressie, 1993] to study seagrass cover drivers, resilience, and recovery in both Moreton Bay. Specifically, we provide and contextualize answers to four specific research questions, applicable to both Moreton Bay:

1. Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?
2. Did seagrass cover improve after the flood event(s)? If so, how and where?

3. Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded? What factors contribute to seagrass resilience?

The spatially explicit statistical models (i.e., spatial models) we used to answer these research questions are related to commonly used linear and generalized linear models but incorporate *spatial dependence*. Spatial dependence measures how similar two sites are based on their distance. Formally, we can build spatial dependence directly into spatial models by leveraging Tobler's First Law of Geography [Tobler, 1970], which states succinctly that "nearby sites tend to be more similar than distant sites". Not only is this intuitive based on "first principles" perspective, but spatial models to provide many advantages for modeling spatial data compared to models that assume independence among observations (i.e., nonspatial models). First, nonspatial models tend to underestimate uncertainty, yielding covariate standard errors and p-values for covariates that are too small and corresponding confidence intervals that are too narrow. Second, spatial models improve prediction at unsampled locations by borrowing strength from nearby observations, yielding more precise predictions and corresponding standard errors that vary by location. Third, spatial models decompose model error into separate spatial and nonspatial components, informing residual and resilience structure in ways that nonspatial models cannot. Fourth, spatial models can provide ecologically meaningful insight into the spatial dependence structure of the process, informing relationships among measured and unmeasured variables as well as future sampling plans. Zimmerman and Ver Hoef [2024] provide a thorough review of spatial models for ecological and environmental data, providing more detail and justification for the claims made above.

Spatial models offer many benefits over commonly used machine learning algorithms like random forests [Breiman, 2001, Cutler et al., 2007, James et al., 2017, Kuhn and Silge, 2022]. First, traditional machine learning algorithms do not incorporate spatial dependence and, by consequence, often perform worse than spatial models at characterizing important covariates and predicting at unobserved locations [Fox et al., 2020]. Second, machine learning algorithms struggle to quantify uncertainty in a principled manner, something spatial models excel at. Third, covariate inference is more straightforward using spatial models, as they naturally provide p-values that enable structured testing of hypotheses. Fourth, spatial models provide a rich set of diagnostics like leverage and Cook's distances [Montgomery et al., 2021], quantities not generally defined for machine learning algorithms. This is not to say that machine learning algorithms cannot be very useful in ecological data analyses, but rather than spatial models are sometimes a more effective tool for modeling spatial data.

Though there are many benefits of spatial models over machine learning algorithms, the reverse is also true. Machine learning algorithms much more simply capture complex nonlinearities "out-of-the-box", bypassing the tuning often required for spatial models that involve nonlinear components like splines. This can be especially useful for ecological data, which are often full of nonlinear relationships [Fox et al., 2017]. Machine learning algorithms also typically rely on few, if any, distributional assumptions and can be more robust against unusual observations like outliers [Hastie et al., 2009]. Machine learning algorithms also tend to be much more computational efficient than spatial models. Importantly, machine learning algorithms and spatial models can be complementary tools that help shed insight into complicated ecological processes (something we reiterate in the upcoming analyses). An exciting and developing area of statistical research involves incorporating spatial modeling strategies directly into machine learning algorithms [Saha et al., 2023, Heaton et al., 2025].

The first spatially explicit model we define is the spatial linear model. The spatial linear model is a *mixed-effects model* [Pinheiro and Bates, 2000, Bolker, 2015] written as

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{1}$$

where $i = 1, 2, ..., n$ indexes the n observations. The quantity y_i is the response variable (for observation i), the β parameters are slope parameters controlling the covariates, $x_{j,i}$ (for $j = 1, 2, ..., p$). The random error τ_i is a spatial random error, and the random error ϵ_i is the nonspatial (i.e., independent) random error. The spatial random error is spatially structured, capturing spatial residual effects like environmental gradients we do not have data to fully represent. The nonspatial random error is not spatially structure and captures nonspatial residual effects like measurement error and microscale variation. The difference between a traditional (i.e., nonspatial) linear model and a spatial linear model is the inclusion of τ_i (in the spatial linear model). The spatial random error implied by τ_i is governed by a *spatial covariance function*. The spatial covariance function allows the model to incorporate spatial dependence, yielding the improved model performance previously described. Figure 1 shows three different spatial covariance functions that describe different ways the spatial covariance decays with distance. Model parameters and standard errors are estimated from data using likelihood-based [Patterson and Thompson, 1971, Harville, 1977, Wolfinger et al., 1994] or semivariogram-based [Cressie, 1985, Curriero and Lele, 1999] estimation methods. Often, the spatial linear model in Equation 1 in is written more compactly using *matrix notation*:

$$\mathbf{y} = \mathbf{X}\beta + \tau + \epsilon, \tag{2}$$

where $\mathbf{X}$ is a matrix that holds the j covariates for each of n observations, and $\mathbf{y}$, β, τ, and τ are vectors that contain each of the n or j quantities from Equation 1. The spatial linear model can be adjusted to account for nonspatial random effects just as one would add random effects to a nonspatial linear mixed model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \tau + \epsilon, \tag{3}$$

where $\mathbf{Z}$ is a random effect design matrix (dimension $n \times k$) that indexes each of the k random effects in $\mathbf{u}$. For more information about specific forms of various spatial covariance functions, see Schabenberger and Gotway [2017] and Zimmerman and Ver Hoef [2024].

Like the traditional linear model, the spatial linear model can capture complex nonlinearities in the data through the use of interaction effects [Faraway, 2002, Lenth and Piaskowski, 2025], splines [Chambers and Hastie, 1992, Ruppert et al., 2003], additive models [Wood, 2017, Pedersen et al., 2019], and penalized regression models like the ridge and lasso [Tibshirani, 1996].

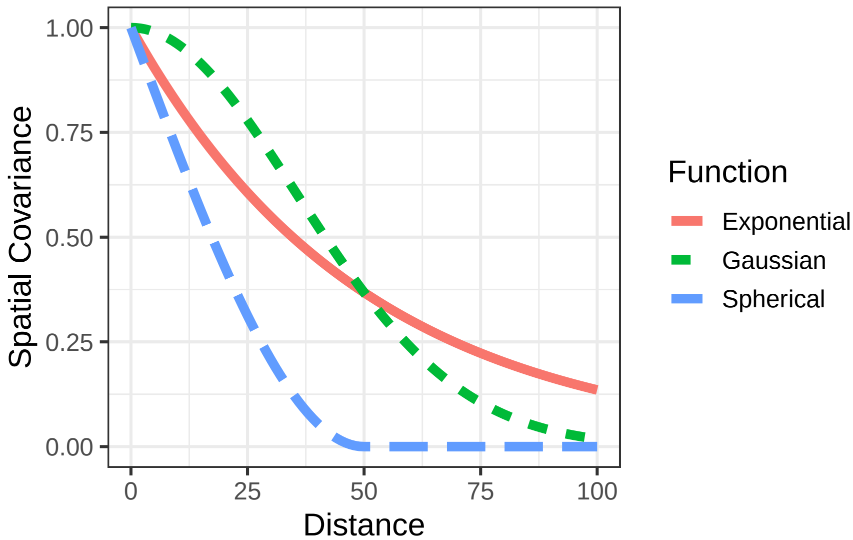

Figure 1: Three spatial covariance functions that represent different spatial dependence behavior.

Generally, this is accomplished by leveraging *basis functions* that break down nonlinear behavior into a set of additive linear components that may include penalty terms which guard against overfitting. Furthermore, there is no assumption that y_i itself is normally distributed; rather, distributional assumptions are placed on the random errors after incorporating the slope parameters. This clarification implies the spatial (and nonspatial) linear models can be quite effective at modeling a range of data that are not normally distributed. For highly nonnormal data, tools like generalized linear models, which we discuss later, can be helpful. Finally, under general conditions, the slope parameters of the spatial (and nonspatial) linear models are approximately normally distributed with a large sample size (see Casella and Berger [2024] for a general justification via the Central Limit Theorem and Zhang and Zimmerman [2005] for nuances regarding spatial data). This is beneficial because it justifies valid hypothesis testing on the slope coefficients (often of primary ecological interest), no matter the distribution of the data (given certain assumptions are satisfied).

One of the most useful applications of the spatial linear model is predicting the response variable at unobserved locations. A very useful spatial prediction method called Kriging has a rich theoretical justification [Cressie, 1990, Stein, 1999]. Kriging is the spatial analogue to best linear unbiased prediction (BLUP) for mixed models [Henderson, 1975]. Via Kriging, the spatial linear model produces BLUPs at each unobserved location of interest, alongside prediction intervals to characterize uncertainty. While useful, sometimes the goal is to create a BLUP for an entire region of interest. This technique, which builds upon the theoretical foundations of Kriging, is called block Kriging [Ver Hoef, 2002, Cressie, 2006]. The intuition is to make point predictions at a very dense grid in the study region and then average these predictions, appropriately characterizing uncertainty. Together, Kriging and block Kriging provide a complete set of tools for spatial prediction, given the fit of a spatial linear model.

Spatial generalized linear models are an extension of spatial linear models to highly nonnormal data [Christensen and Waagepetersen, 2002, Zhang, 2002, Diggle and Ribeiro Jr, 2007, Bonat and Ribeiro Jr, 2016]. Ver Hoef et al. [2024] proposed a novel application of the Laplace approximation to enable restricted maximum likelihood estimation of spatial generalized linear models. Their model formulation builds upon Equation 1 and is given by

$$g(\mu_i) = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{4}$$

where $g(\mu_i)$ is a "link function" that links the distribution of y_i to a linear function of its mean, μ_i. Common link functions include the log link for count and skewed data and the logit link for binomial and proportion data (Table 1). Using the same argument as for spatial linear models, nonspatial random effects can be added to spatial generalized linear models.

Table 1: Common generalized linear model families, link functions, and appropriate data types.

Family	Link Function	Link Name	Data Type
Binomial	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Binary; Binary Count
Poisson	$f(\mu) = \log(\mu)$	Log	Count
Negative Binomial	$f(\mu) = \log(\mu)$	Log	Count
Beta	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Proportion
Gamma	$f(\mu) = \log(\mu)$	Log	Skewed
Inverse Gaussian	$f(\mu) = \log(\mu)$	Log	Skewed

Spatial models are complicated, both theoretically and computationally. This has contributed to their relative lack of use in the ecological community. However, recent advances in open-source software have made spatial models more accessible to ecologists who have strong subject matter expertise but may lack sufficient statistical background or computational skills to implement them from scratch. One such advancement is the `spmodel` **R** package [Dumelle et al., 2023], which emulates base-**R** functions like `lm()` and `glm()` to account for spatial dependence via `splm()` and `spglm()`, respectively. Familiar **R** functions like `summary()` and `predict()` can be used directly on models fit via `spmodel`. Importantly, `spmodel` also has tools for spatial random forest residual modeling [Fox et al., 2020], applications to large data of several thousand observations via spatial indexing [Ver Hoef et al., 2023], and cross validation [Stone, 1974, Roberts et al., 2017].

3 The Data

We started with just over 10,000 seagrass cover observations in Moreton Bay across three financial years (2020-2021, 2022-2023, and 2023-2024). eReefs data [Steven et al., 2019] with modeled water quality covariates related to turbidity and salinity were only available at 7,657 observations. Because one of our research questions involved connecting water quality covariates to management actions, we proceed with this subset of 7,657 observations for modeling. Of the 7,657 observations, 1,995 were in 2020-2021 (pre-flood), 3,329 were in 2022-2023, and 2,333 were in 2023-2024 (Table 2).

Table 2: Total seagrass observations in Moreton Bay by financial year.

Financial Year	Sample Size	Percent (of Total)
2020-2021	1,995	26.1%
2022-2023	3,329	43.5%
2023-2024	2,333	30.5%
Total	7,657	100.0%

4 Water Quality and Geographic Modeling

Total cover is a percentage that ranges from 0 (no seagrass) to 100 (seagrass entirely contained in an area). Figure 2 shows total cover by financial year throughout the Bay. Total cover tends to be largest along the eastern shoreline of the Bay and lowest in the middle of the Bay. Figure 2 also reveals that 2022-2023 had by far the most sampling in the middle of the Bay.

We considered for modeling several water quality and geographic covariates. Table 3 summarize the water quality variables, which included both ambient and flood-related metrics related to turbidity, salinity, and temperature. The “Variable” column indicates the water quality variable turbidity, salinity, or temperature. The “Flood” variable indicates whether the measurement corresponds to a flood variable or not. The “Period” variable indicates the prior period (in days) relative to sampling or indicates whether the variable corresponds to a flood event. The “Type” column is the type of statistic, either a mean, minimum, maximum, number of days above (or below) a cutoff during “Period”, or a maximum spell (i.e., duration) above (or below) a cutoff during “Period”. The “Included” variable indicates whether the variable was included as a covariate in the final spatial model. We included only a few variables in the final model: mean turbidity in the 90 days prior to sampling, mean turbidity during the flood event, and mean temperature in the 90 days prior to sampling. We only included a single turbidity metric prior to sampling because these variables were all quite correlated with one another, making it hard for the model to tease apart signals. We believed 90 day mean turbidity prior to sampling to be the most accurate “running average” of turbidity prior to sampling. We used a similar rationale to select only a single turbidity metric from the flood and a single temperature metric prior to sampling. We omitted a flood salinity metric primarily because we lacked data for salinity prior to sampling, thus it would be challenging to establish a “baseline” for salinity in order to tease apart prior to sampling effects from flood

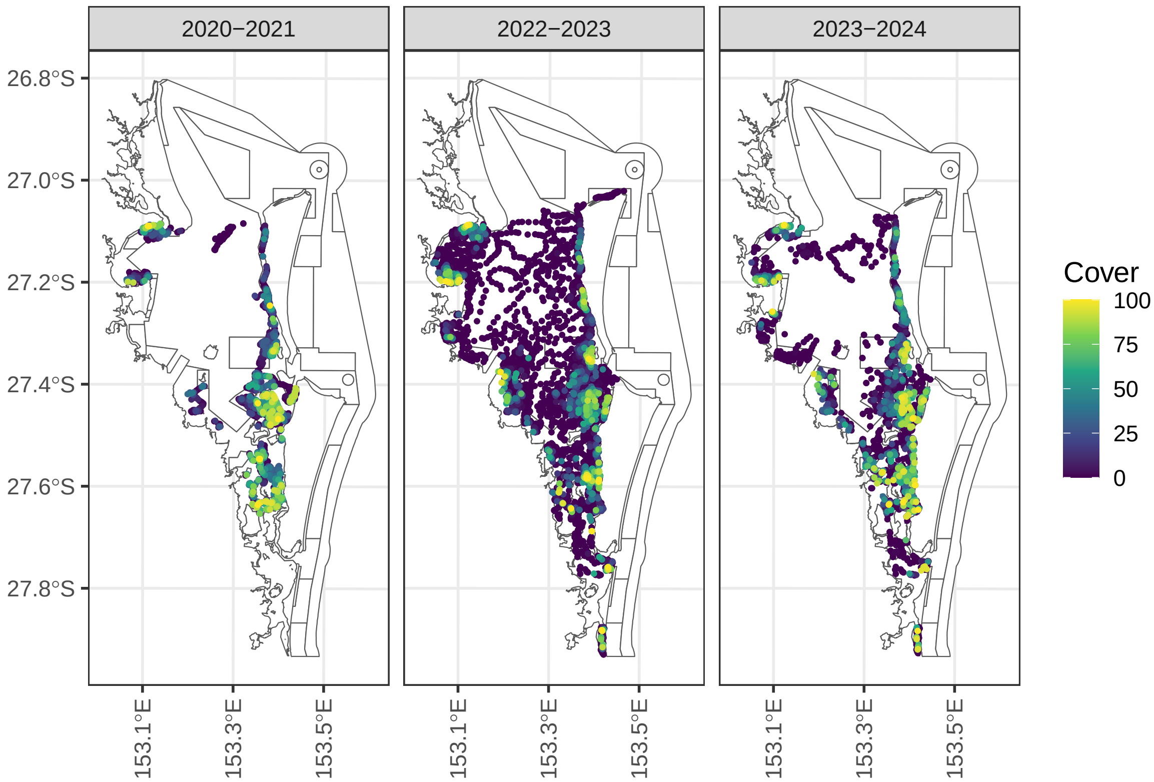

Figure 2: Total seagrass cover in Moreton Bay by financial year.

effects (moreover, they did not seem strongly related to total cover). We provide more detail regarding the modeling in Section 4.1.

Table 3: Moreton Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Turbidity (NTU)	No	7	Mean	NA	No
Turbidity (NTU)	No	30	Mean	NA	No
Turbidity (NTU)	No	90	Mean	NA	Yes
Turbidity (NTU)	No	7	Above	3	No
Turbidity (NTU)	No	30	Above	3	No
Turbidity (NTU)	No	90	Above	3	No
Turbidity (NTU)	No	7	Above	5	No
Turbidity (NTU)	No	30	Above	5	No
Turbidity (NTU)	No	90	Above	5	No
Turbidity (NTU)	No	7	Above	6	No
Turbidity (NTU)	No	30	Above	6	No
Turbidity (NTU)	No	90	Above	6	No
Turbidity (NTU)	Yes	Flood	Above	3	No
Turbidity (NTU)	Yes	Flood	Above	5	No
Turbidity (NTU)	Yes	Flood	Above	6	No
Turbidity (NTU)	Yes	Flood	Duration	3	No
Turbidity (NTU)	Yes	Flood	Duration	5	No
Turbidity (NTU)	Yes	Flood	Duration	6	No
Turbidity (NTU)	Yes	Flood	Mean	NA	Yes
Turbidity (NTU)	Yes	Flood	Max	NA	No
Salinity (PSU)	No	7	Below	20	No
Salinity (PSU)	No	30	Below	20	No
Salinity (PSU)	No	90	Below	20	No
Salinity (PSU)	No	7	Below	25	No
Salinity (PSU)	No	30	Below	25	No
Salinity (PSU)	No	90	Below	25	No
Salinity (PSU)	No	7	Below	30	No
Salinity (PSU)	No	30	Below	30	No
Salinity (PSU)	No	90	Below	30	No
Salinity (PSU)	Yes	Flood	Below	20	No
Salinity (PSU)	Yes	Flood	Below	25	No
Salinity (PSU)	Yes	Flood	Below	30	No
Salinity (PSU)	Yes	Flood	Duration	20	No
Salinity (PSU)	Yes	Flood	Duration	25	No
Salinity (PSU)	Yes	Flood	Duration	30	No
Salinity (PSU)	Yes	Flood	Mean	NA	No

Table 3: Moreton Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Salinity (PSU)	Yes	Flood	Min	NA	No
Temperature (deg C)	No	7	Mean	NA	No
Temperature (deg C)	No	30	Mean	NA	No
Temperature (deg C)	No	90	Mean	NA	Yes

Table 4 summarizes the geographic variables, which have the following structure:

- The "Sediment Type" variable has six levels: Sand, Sandy Mud, Muddy Sand, Mud, Rock, and Other.
- The "Tidal Range" variable has three levels: Shallow Subtidal, Intertidal, and Deep Subtidal.
- The "Years Since Flood" is numeric from zero (pre flood) to 1.93 years post flood (February, 2024).
- The "Nearest River" variable has six levels: Brisbane River, Caboolture River, Logan River, Ningi Creek, Pimpama Coomera and Nerang Rivers, and Pine River.
- The "Distance to (Nearest River) Outlet (in km)" variable is numeric from 0.24 to 34.97.
- The "Growing Season" variable has two levels: Growing (September - December) and Not Growing (January - August).
- The EHMP subzone variable has eleven levels: Bramble Bay, Broadwater, Central Bay, Deception Bay, Eastern Banks, Moreton Bay Open Coastal Waters, North-Eastern Moreton Bay, Northern Banks Open Coastal Waters, South-Eastern Moreton Bay, Southern Bay, and Waterloo Bay (Figure 3).

Table 4: Moreton Bay geographic variables considered for modeling.

Variable	Effect	Included
Sediment Type	Fixed	Yes
Tidal Range	Fixed	Yes
Years Since Flood	Fixed	Yes
Nearest River	Fixed	No
Distance to Outlet	Fixed	No
Growing Season	Fixed	No
EHMP Subzone	Random	Yes

From Table 4, we kept the sediment type, tidal range, and years since flood variables as covariates in the model. The nearest river and distance to nearest outlet variables were removed, as they did not seem related to total cover. We removed growing season from the model after determining that it likely is not ecologically useful, as a simple month proxy likely does not adequately account for the complexities of the more nuanced aspects of seagrass growth with time. We used EHMP subzone as a random effect to account for within-subzone correlation that may impact the relationships among total cover and the remaining variables. We provide more detail regarding the statistical modeling in Section 4.1.

4.1 Model Fit

Table 5: Variables in the final Moreton Bay total cover model, their type (numeric or categorical), and if categorical, the number of unique levels.

Variable	Type	No. Levels
90 Day Temperature	Numeric	NA
90 Day Turbidity	Numeric	NA
Mean Flood Turbidity/Years Since Flood^2	Numeric	NA
Sediment Type	Categorical	6
Tidal Range	Categorical	3

To describe total seagrass cover in Moreton Bay, our final model was a spatial linear model (Equation 1) with the following explanatory variables: tidal range, sediment type, 90 day mean temperature (prior to sampling), pre- vs post-flood, years since the flood, 90 day turbidity (prior to sampling), and mean turbidity during the flood period scaled by time since the flood squared (Table 5). We scaled mean turbidity during the flood period so that the effect of turbidity during the flood event was allowed to decay with time since the flood. Intuitively, this means that with time since the flood, the impact of turbidity during the flood event is lessened. We also considered scaling by time since flood (not squared), which produced similar results. Furthermore, we used an exponential spatial covariance function and included a random effect for EHMP subzone. We fit the model using restricted maximum likelihood via the `splm()` function in `spmodel`.

Table 6: ANOVA table of covariates in the Moreton Bay water quality and geographic model.

Effects	df	Chi-sq	p.value
Tidal Range	2	608.900	0.000
Sediment Type	5	113.720	0.000
90 Day Temperature	1	16.967	0.000
Mean Flood Turbidity/Years Since Flood^2	1	11.966	0.001
90 Day Turbidity	1	7.854	0.005

Table 6 shows the results from an analysis of variance applied to the fitted model. All explanatory variables had large test statistics and small p-values (p-value < 0.01); here are some relevant takeaways:

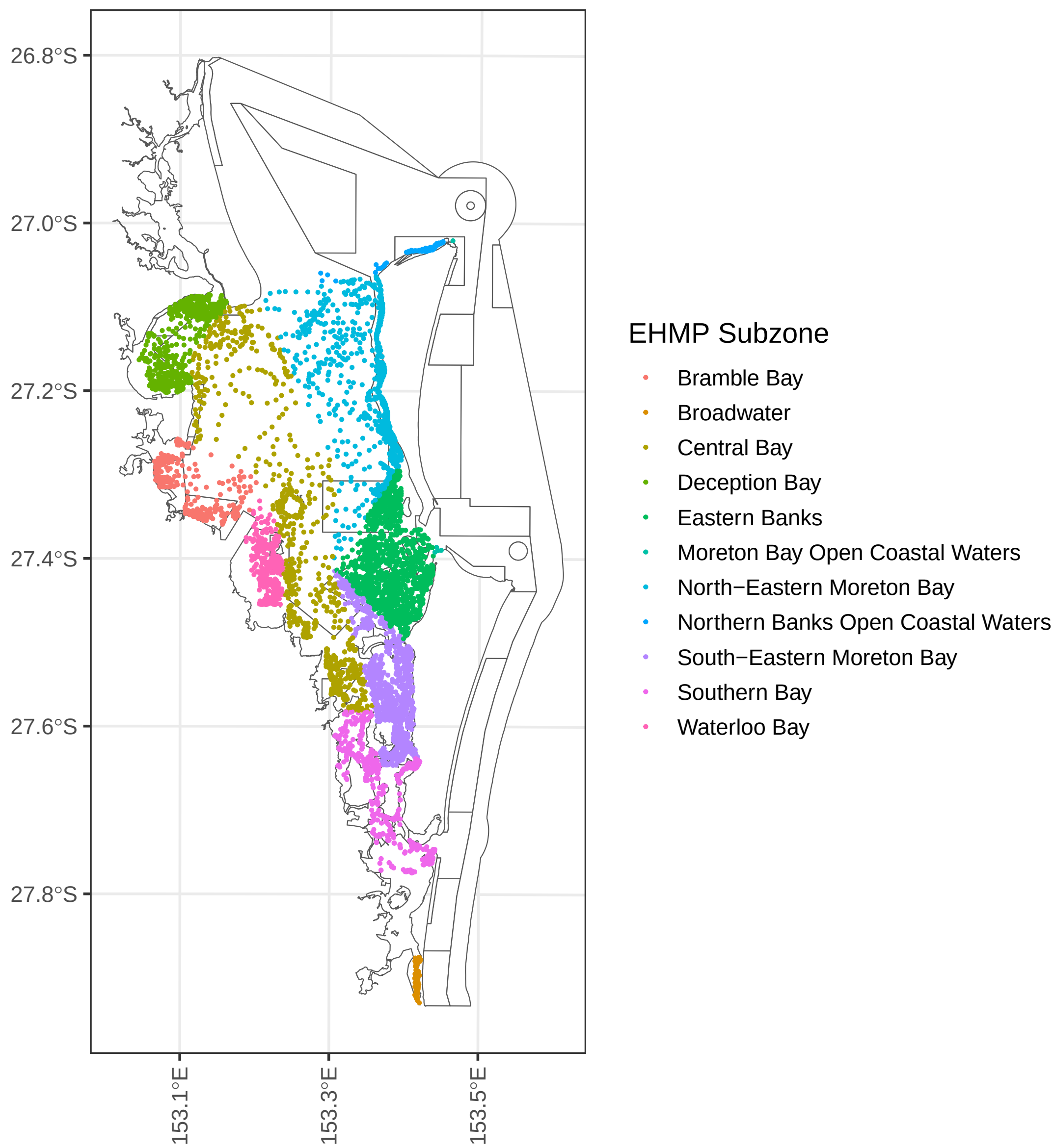

Figure 3: Observations and their EHMP Subzones in Moreton Bay.

- Tidal Range: The estimated (marginal) average total cover is largest in the intertidal (26.89%), followed by shallow subtidal (16.51%) and deep subtidal (4.88%).
- Sediment Type: The estimated (marginal) average total cover is largest in the other (25.6%), followed by muddy sand (19.1%), sandy mud (17.8%), mud (14.3%), sand (14.2%), and rock (5.62%).
- 90 Day Temperature: A one-degree (C) increase in 90 day temperature is associated with a decrease in average total cover by 0.88% points.
- Mean Flood Turbidity/(Years Since Flood)^2: An one-unit increase in mean turbidity during the flood is associated with a decrease in average total cover by 2.05% points/(Years Since Flood)^2 (Figure 4).
- 90 Day Turbidity: A one-unit increase in 90 day turbidity is associated with a decrease in average total cover by 7.18% points (fig-mb_turb_90d_fld).

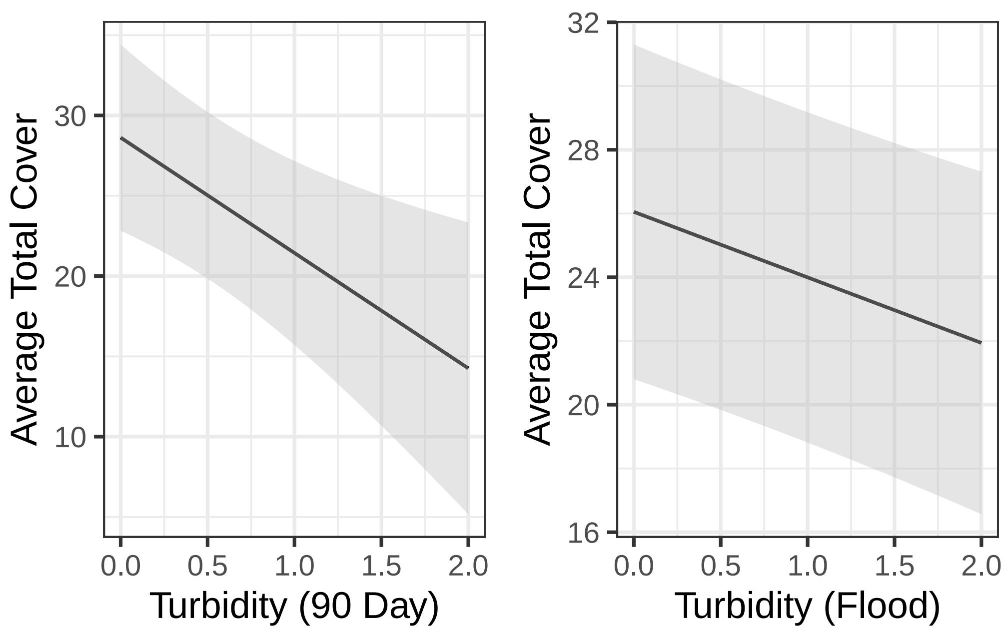

Figure 4: Average total cover as a function of 90 day mean turbidity (left) and flood turbidity (right). Estimates (and confidence intervals) were evaluated at one year post flood, at the means of all other numeric variables, and for the intertidal range and Sand sediment type (the same trends occur for both variables among the remaining species and subzones).

Together, these results can inform triggers for management responses in both ambient and flood conditions. For example, given the decrease in total cover with turbidity, one could propose a specific turbidity threshold above which management intervention is warranted. Note that the estimated marginal means for tidal range and sediment type were calculated using the `emmeans` **R** package [Lenth and Piaskowski, 2025].

We may inspect model diagnostics to better understand the overall model fit. Here we discuss how to partition the model into different components of variability, infer decay behavior of spatial autocorrlation, analyze standardized residuals to assess model assumptions, and identify influential observations that have a sizable impact on model fit. From Table 7, we see that of the total model variability, 11.2% is attributable to the explanatory variables (this quantity is sometimes called the Pseudo R-squared), 31.6% is attributable to the spatial random error, 7.3% is attributable to the EHMP subzone, and the remaining variability is independent random error. The estimated exponential spatial range (ϕ) is 745.5 meters, which suggests two total cover observations are approximately (spatially) uncorrelated at a distance of $3\phi \approx 2$km, or two kilometers. While the raw residuals will inform our resilience modeling in Section 5, we can check certain model assumptions using the standardized residuals, which have been decorrelated. Generally, the standardized residuals are centered around zero, though there is a slight right skew suggesting some observed total cover values are larger than expected (according to the model). Observations are called "influential" if they highly influence model fit. Influence is often measured via Cook's distance, and no observations exceed the often-used Cook's distance threshold value of one [Cook and Weisberg, 1982].

Table 7: Variance components in the Moreton Bay water quality and geographic model.

Variance Component	Percent (of Total Variability)	Parameter Value
Covariates	11.2%	NA
Spatial Effect	31.6%	182.1
Independent Error	49.9%	287.5
EHMP Subzone	7.3%	745.5

4.2 Block Kriging

A tremendous benefit of using the spatial linear model is that block Kriging can be directly applied to predict the mean total seagrass cover bay-wide for any combination of explanatory variables and geographic regions, even if these variables were not directly used in modeling. Figure 5 and Figure 6 show that there are several EHMP subzones and conservation zones that have recovered to pre-flood levels (e.g., Deception Bay, Marine National Park Zone), while others have not (e.g., Southern Bay, General Use Zone). While EHMP subzone was included in the model as a random effect, conservation zone type was not – via block Kriging, we can still quantify broad trends in total seagrass cover. Figure 7 shows that total seagrass cover is recovering less effectively in the intertidal range than the shallow or deep subtidal ranges, while Figure 8 shows that total seagrass cover is recovering bay-wide since the flood. Another useful application of block Kriging is scenario modeling, which can be used to characterize the impact of various future changes to broad-scale climatic patterns.

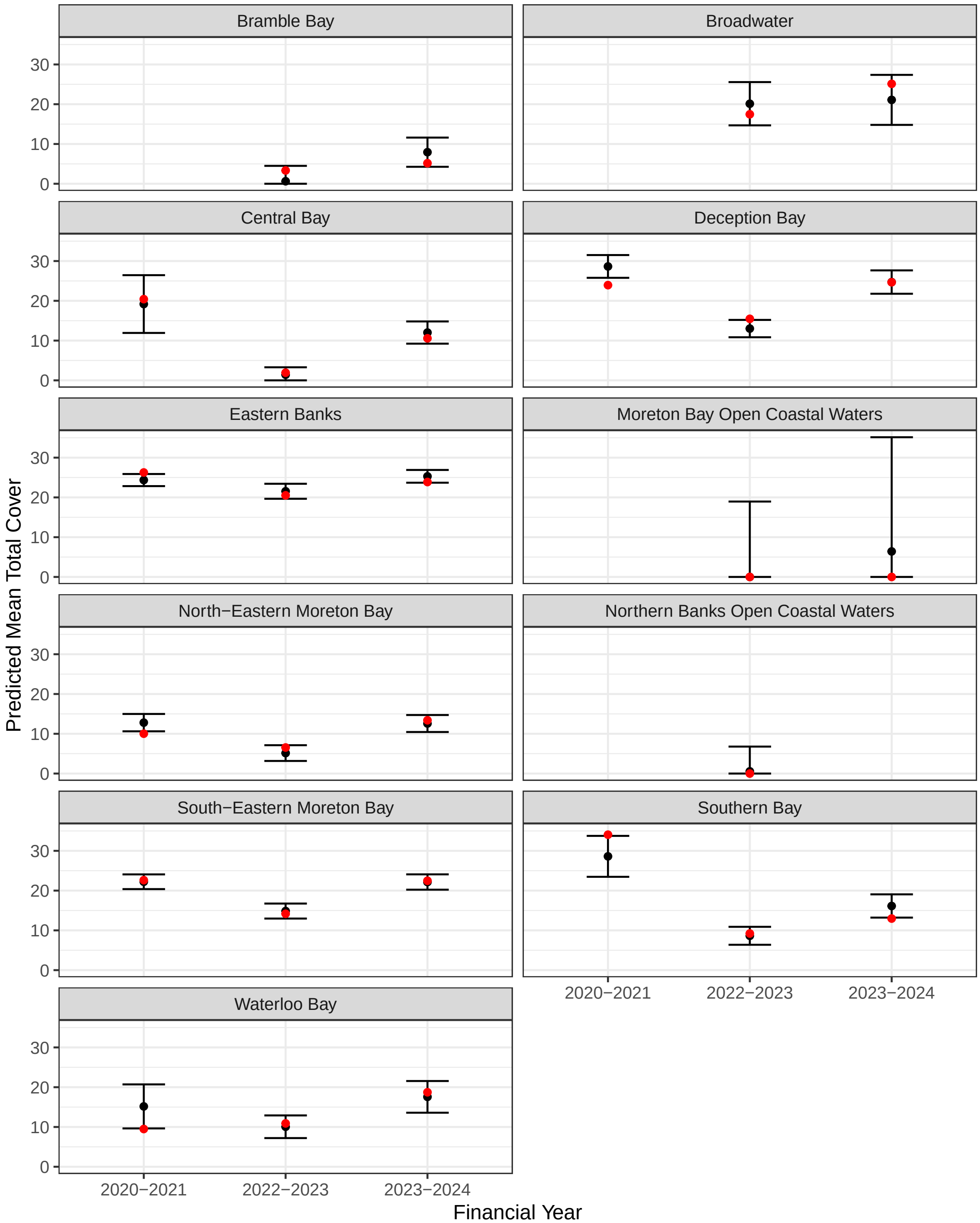

Figure 5: Average total cover predictions by EHMP subzone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

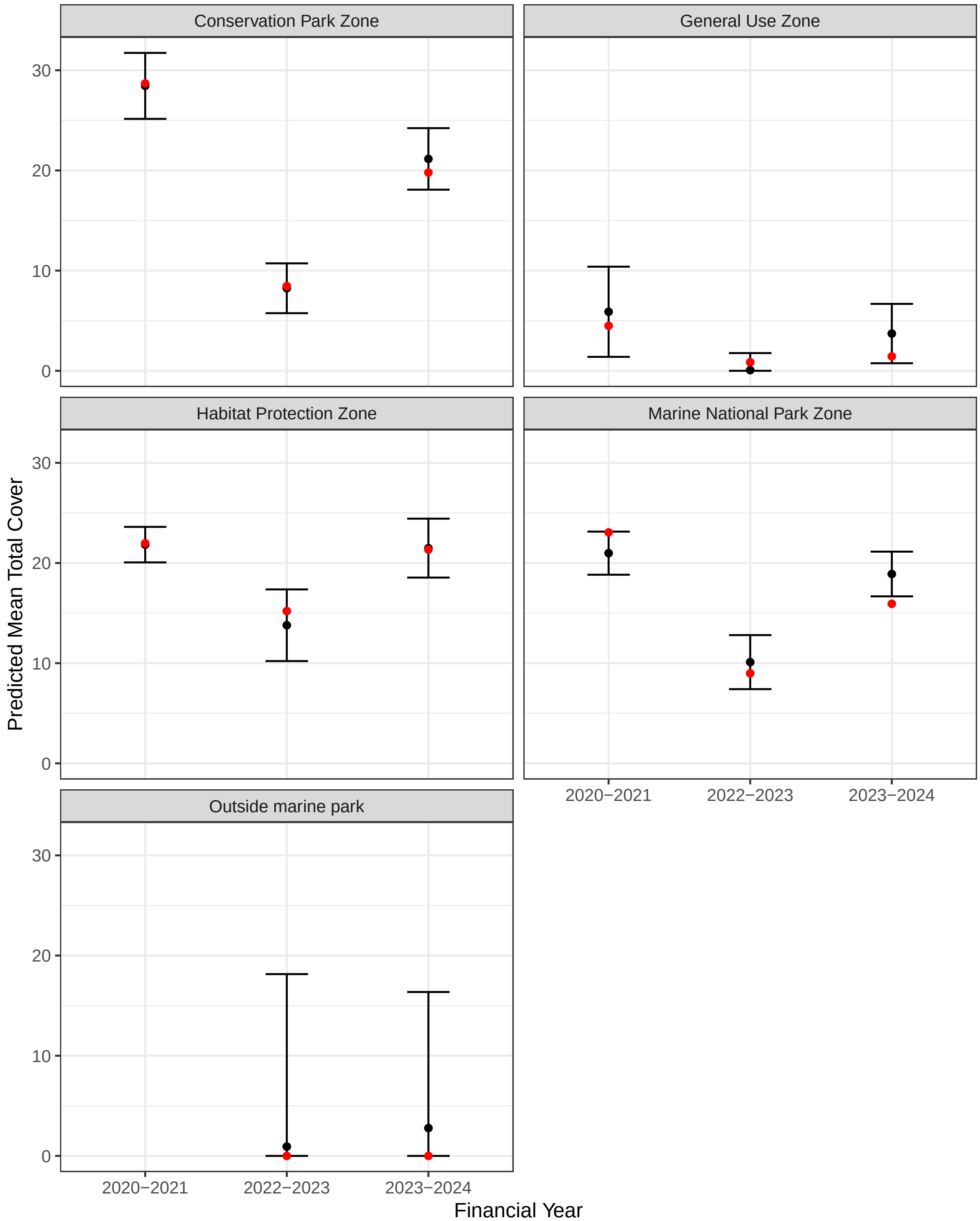

Figure 6: Average total cover predictions by conservation zone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

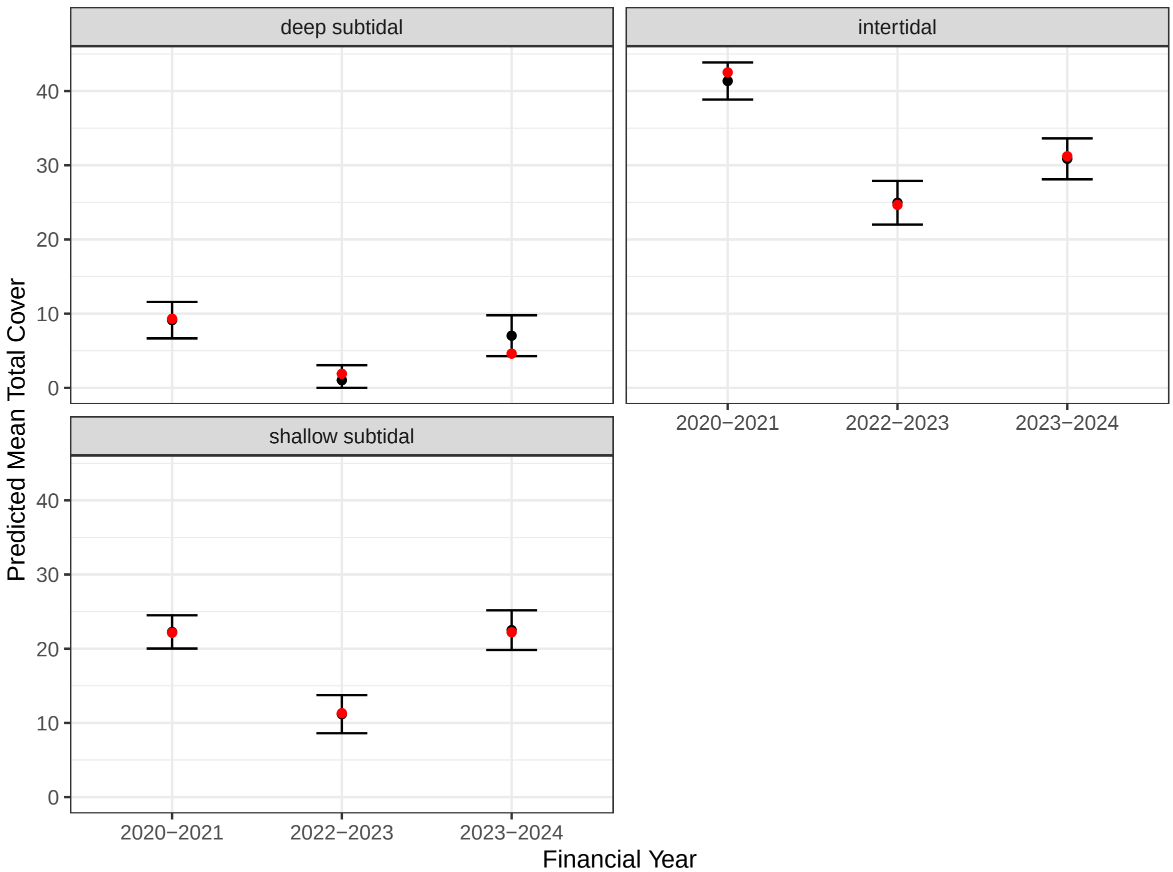

Figure 7: Average total cover predictions by tidal range. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

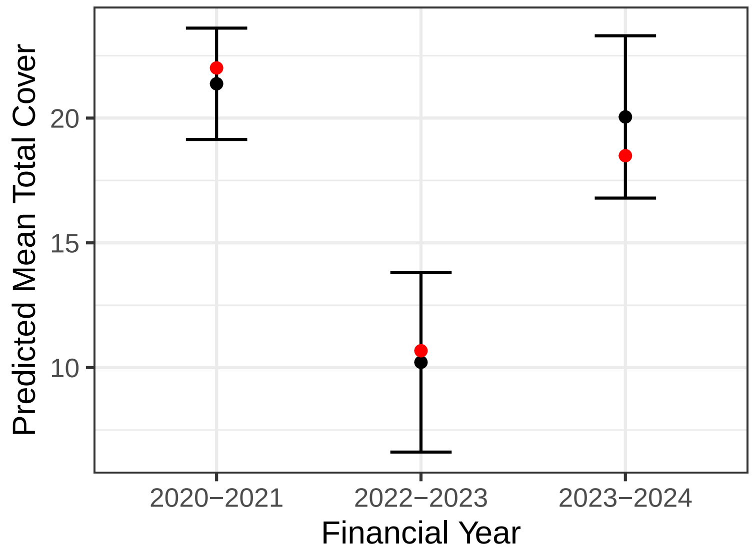

Figure 8: Average total cover predictions in Moreton Bay. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

4.3 Alternative Approaches

In addition to the spatial linear model, we considered several different modeling approaches (all using the same explanatory variables): a nonspatial linear model, a spatial beta regression model, a spatial gamma regression model, and a random forest model. The nonspatial linear model provided a useful baseline to compare the spatial linear model against, elucidating the impact of spatial dependence on model fit. The spatial beta regression model is a generalized linear model used to characterize proportion data between zero and one. To adhere to this restriction, total cover was 1) scaled to a proportion and 2) if exactly zero (one), set to 0.01 (0.99). The spatial gamma regression model is a generalized linear model used to characterize skewed positive data. To adhere to this restriction, if total cover was exactly zero, it was set to 0.01. Finally, the random forest provided a useful comparison to a machine learning approach, quite different than the aforementioned statistical approaches. We characterized the directionality of random forest effects using partial dependence plots (Figure 9).

Importantly, all five approaches found important relationships among the explanatory variables and total cover, in the directions expected. From Table 8, the spatial linear model and random forest had the lowest out-of-sample root-mean-squared-prediction error, followed by the generalized linear models and then by the nonspatial linear model. Random forest partial dependence plots [Greenwell, 2017], which characterize the marginal behavior of a variable in a random forest model, suggested linear effects of the model variables to be reasonable. Given this context, we chose the spatial linear model as the final model.

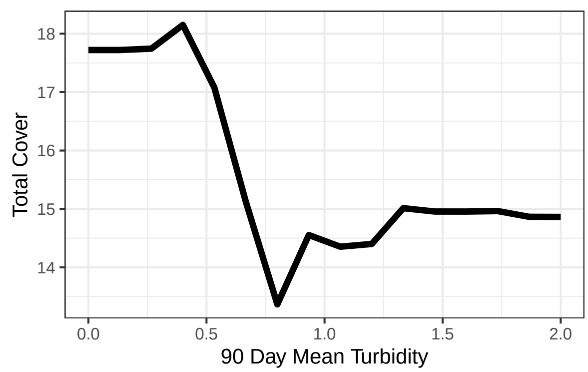

Figure 9: Random forest partial dependence for total cover as a function of 90 day mean turbidity.

Table 8: Moreton Bay out of sample prediction metrics for the various approaches. Mean bias (MBias) tended to be small relative to root-mean-squared-prediction error (RMSPE).

Approach	MBias	RMSPE
SPLM	-0.02	18.41
RF	-0.11	18.84
SPGLM-Beta	-2.76	19.60
SPGLM-Gamma	5.42	20.07
LM	0.00	22.54

5 Resilience Modeling

The model described in Section 4 quantifies, on a bay-wide scale, the effects of various water quality and geographic drivers on total seagrass cover. It does not, however, quantity resilience on a site-specific scale. Gilby et al. [2023] provide a framework to quantify resilience on a site-specific scale. They suggest identifying hotspots (i.e., brightspots), sites that are performing better than expected (according to some model fit), as well as coldspots, sites that are performing worse than expected (according to some model fit). They identify hotspots and coldspots based on deviations exceeding expected bounds (based on confidence intervals) based on the model fit. They argue these hotspots and coldspots provide significant utility, both in better understanding the drivers of site-specific resilience as well as informing subsequent management action.

Table 9: Moreton Bay resilience variables considered for modeling.

Variable	Effect	Included	Nonlinearity Degree
Coral Habitat Total Area	Fixed	Yes	2
Seagrass Habitat Total Area	Fixed	Yes	2
Mangrove Habitat Total Area	Fixed	Yes	2
Gastropod Habitat Total Area	Fixed	Yes	2
Distance to Coral Habitat	Fixed	Yes	3
Distance to Seagrass Habitat	Fixed	Yes	2
Distance to Mangrove Habitat	Fixed	Yes	2
Distance to Gastropod Habitat	Fixed	Yes	2
Species Richness	Fixed	Yes	1
Habitat Count	Fixed	Yes	2

Table 9: Moreton Bay resilience variables considered for modeling.

Variable	Effect	Included	Nonlinearity Degree
EHMP Subzone	Random	Yes	NA

Building from Gilby et al. [2023], we use the residuals from the model in Section 4 as the basis for studying site-specific resilience, linking the values of these residuals to site-specific variables like species habitat, species richness, and bathymetry via a separate spatial linear model, with a random effect for EHMP subzone. These models capture varying nonlinear behavior in the residuals as a function of the resilience variables using B-spline basis functions [de Boor, 1978, Eilers and Marx, 1996] whose degree is informed by random forest partial dependence plots (built from a random forest model on the residuals). While Gilby et al. [2023] focused more on specific hotspots, linking environmental variables to those sites, we used our framework to provide a different set of information – the characterizing of the most important drivers of resilience, on average. Because were were interested in hypotheses not parsimony, we included all potential variables as fixed effects in the model. These included area of and distance to coral, seagrass, mangrove, and gastropod habitat, species richness and dominant species, habitat count, and bathymetry metrics (Table 9), which have the following structure:

- The "Coral Habitat Total Area" is numeric from 0 msq to 616,180 msq.
- The "Seagrass Habitat Total Area" is numeric from 0 msq to 773,161 msq.
- The "Mangrove Habitat Total Area" is numeric from 0 msq to 561.317 msq.
- The "Gastropod Habitat Total Area" is numeric from 0 msq to 114,795 msq.
- The "Distance to Coral Habitat" is numeric from 0 m to 35,387 m.
- The "Distance to Seagrass Habitat" is numeric from 0 m to 11,382 m.
- The "Distance to Mangrove Habitat" is numeric from 0 m to 27,617 m.
- The "Distance to Gastropod Habitat" is numeric from 0 m to 32,034 m.
- The "Species Richness" is numeric (integer) from 0 unique species to 4 unique species
- The "Habitat Count" is numeric (integer) from 0 unique habitats to 4 unique habitats.
- The "Dominant Species" has seven levels: "Absent", "CS", "HD", "HO", "HS", "SI", and "ZMHU".
- The "Bathymetry Mean Depth" is numeric from -35.95 m to 0.79 m.
- The "Bathymetry Mean Curvature" is numeric from -0.002 degrees to 0.0001 degrees.
- The "Bathymetry Mean Slope" is numeric from 0.009 degrees to 4.27 degrees.
- The "EHMP subzone" variable has eleven levels: Bramble Bay, Broadwater, Central Bay, Deception Bay, Eastern Banks, Moreton Bay Open Coastal Waters, North-Eastern Moreton Bay, Northern Banks Open Coastal Waters, South-Eastern Moreton Bay, Southern Bay, and Waterloo Bay (Figure 3).

5.1 Model Fit

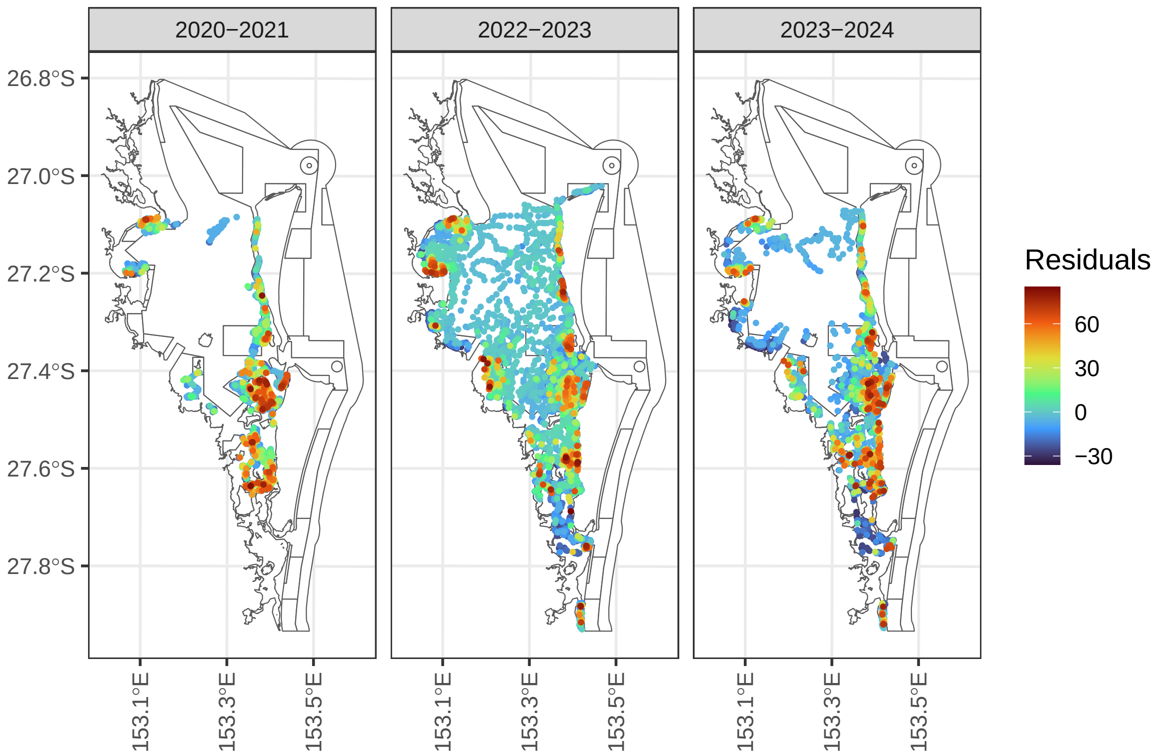

Figure 10: Residuals for reslience analysis in Moreton Bay by financial year.

Table 10: ANOVA table of covariates in the Moreton Bay resilience model.

Effects	df	Chi-sq	p.value
Dominant Species	6	1637.726	0.000
Species Richness	1	22.188	0.000
Bathymetry Mean Depth	2	21.876	0.000
Bathymetry Mean Curvature	2	8.999	0.011
Gastopod Habitat Total Area	2	5.155	0.076
Bathymetry Mean Slope	2	4.768	0.092
Coral Habitat Total Area	2	2.845	0.241
Seagrass Habitat Total Area	2	1.971	0.373
Distance to Mangrove Habitat	2	1.871	0.392
Distance to Seagrass Habitat	2	1.836	0.399
Habitat Count	2	0.722	0.697
Distance to Gastopod Habitat	2	0.613	0.736
Distance to Mangrove Habitat	2	0.536	0.765
Distance to Coral Habitat	3	0.718	0.869

From Table 10, we find strong evidence (p-value < 0.001) that dominant species, species richness, and bathymetry mean depth are associated with resilience:

- Dominant Species: The estimated (marginal) average resilience score is largest for CS (38.4), followed by ZMHU (18.8), HS (10.8), SI (10.7), HO (3.7), and HD (-3.1).
- Species Richness: An increase of species richness by one species was associated with an average increase in resilience by approximately two points (Figure 11).
- Bathymetry Mean Depth: Deeper sites tended to be more resilient than shallower sites, on average (Figure 11).

We found moderate evidence (p-value < 0.1) that bathymetry mean curvature, gastropod habitat total area, and bathymetry mean slope are associated with resilience:

- Bathymetry Mean Curvature: Sites with more curvature tended to be less resilient than sites with less curvature.
- Gastropod Habitat Total Area: Sites with more gastropod habitat area tended to be more resilient than sites with less gastropod habitat area.
- Bathymetry Mean Curvature: Sites with less extreme slopes (between zero and one) tended to have increasing resilience with slope, while sites with more extreme slopes (greater than one) tended to have decreasing resilience with slope.

We found little evidence (p-value > 0.1) the remaining variables were associated with resilience (after controlling for other modeled variables).

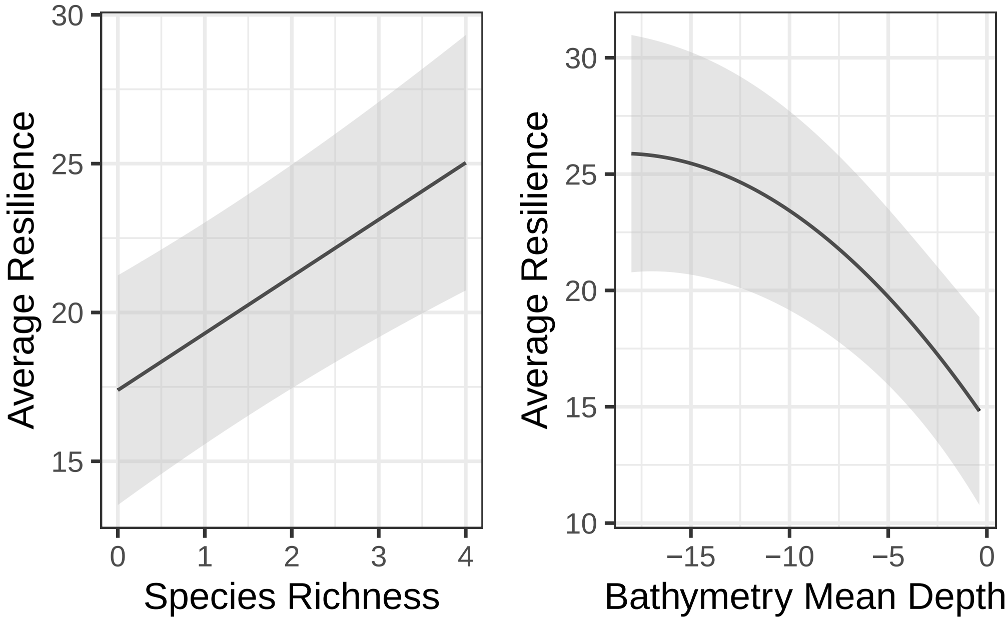

Figure 11: Average species richness (left) and bathymetry mean depth (right) effects on resilience. Predictions (and confidence intervals) were evaluated at the means of all other numeric variables and for the ZHMU species in the Southern Banks (similar trends occur for both variables among the remaining species and subzones).

5.2 Alternative Approaches

Another way we studied resilience was using a random forest (fit to the same residuals). Largely, we found similar results as the spatial linear model. Notably, for the random forest:

- Dominant species was most important, with a similar ranking of resilient species as for the spatial linear model approach (e.g., CS, ZMHU most resilient and Absent least resilient.
- Species richness was second-most important, with a general increasing trend in resilience with richness.
- Distance to gastropod habitat was third-most important, with a decreasing trend with at small distances (less than 15,000 meters) and an increasing trend at large distances (greater than 15,000 meters)
- Bathymetry mean distance was fourth-most important, with a general decreasing trend with decreasing depth Figure 12, similar to the spatial linear model.
- Distance to gastropod habitat, distance to mangrove habitat, distance to coral habitat, and seagrass area had similar importance as the remaining bathymetry variables
- The remaining variables (distance to seagrass habitat, mangrove area, EHMP subzone, habitat count, coral area, and gastropod area) were least important.

Other alternatives to explore involve using the residuals from water quality and geographic models fit using the spatial generalized linear models and random forests. These residuals could be analyzed using spatial linear models (recall that the residuals are unconstrained) and random forests. We expect these alternative approaches to yield generally similar results, however.

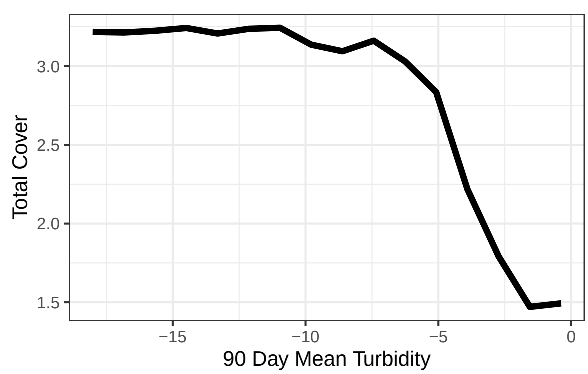

Figure 12: Random forest partial dependence for total cover as a function of bathymetry mean depth.

6 Limitations and Future Suggestions

While the spatial linear model was useful for characterizing broad patterns related to tidal range, sediment type, temperature, and turbidity and site-specific patterns capturing resilience, all while accounting for spatial dependence, there are some limitations. First, the eReefs water quality data are modeled, not directly observed in the field, potentially propagating modeling error. Second, there are many correlated water quality variables, making it hard (for any model) to isolate the effects of each component of water quality pre- and post-flood. Third, there is a lack of baseline mean salinity in the 90 days prior to sampling, which makes it challenging to directly study the effect of mean salinity during the flood period. Fourth, there is a lack of data immediately after the flood – the first post-flood sampling effort occurred four months after the flood. Fifth, there is a lack of addition data sources that could help quantify the impact of various explanatory variables on total pre and post-flood, potentially varying across both space and time (e.g., light attenuation, biological interactions, etc.). Sixth, the locations at which data were collected were not randomly sampled. Randomly sampling sites helps to guard against sampling preferentially [Diggle et al., 2010] and provides more robust and representative data sources [Dumelle et al., 2022].

7 Revisiting the Research Questions

Research Question 1: Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?

We identified ambient turbidity, flood turbidity, and temperature as significant water quality triggers for seagrass cover responses. Ambient and flood-related turbidity act through distinct mechanisms and at different magnitudes — a one-unit increase in 90-day mean turbidity was associated with a 7.18 percentage point decrease in total cover, while the effect of flood turbidity decayed with time since the flood. Across tidal ranges, intertidal sites supported the highest average total cover (26.89%), followed by shallow subtidal (16.51%) and deep subtidal (4.88%), suggesting that management thresholds may need to be calibrated differently depending on depth zone, as deeper sites with inherently lower cover may be disproportionately sensitive to turbidity increases. Similarly, sediment type affected baseline cover levels, as sites with muddy sand and sandy mud substrates supported higher average cover than sand or rock sites. Together, these results suggest that management triggers should differ depending on whether the driver is chronic ambient turbidity or an acute flood event, and should further account for the tidal and sediment context of the sites in question (e.g., perhaps turbidity-driven management responses are most important in areas of densely populated seagrass).

Research Question 2: Did seagrass cover improve after the flood event(s)? If so, how and where?

Total seagrass cover improved bay-wide between 2022–2023 and 2023–2024, suggesting the system has been recovering since the flood. However, recovery was spatially heterogeneous — Deception Bay and the Marine National Park Zone returned to pre-flood cover levels, while Southern Bay and the General Use Zone lagged considerably behind. Recovery was also less effective in the intertidal tidal range compared to shallow and deep subtidal ranges, suggesting that intertidal sites may warrant prioritized management attention in the near term.

Research Question 3: Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded? What factors contribute to seagrass resilience?

We identified several site-specific indicators of seagrass resilience. Dominant species identity, species richness, and bathymetry mean depth were the strongest predictors of resilience, with sites dominated by CS or ZMHU species, higher species richness, and greater water depth tending to outperform model expectations and hence, infer greater resilience. Gastropod habitat area, bathymetric curvature, and slope provided moderate additional explanatory power. These findings suggest that management actions aimed at protecting structurally diverse, species-rich meadows at deeper sites are likely to be most resilient, while sites with low species richness dominated by other species (or no seagrass) may require more active intervention to support recovery.

8 Discussion

This report applies a rigorous, spatially explicit statistical framework to characterize the drivers, recovery, and resilience of seagrass cover in Moreton Bay. The analytical approach taken here is well-suited to the structure and scale of the data, and the findings have direct implications for adaptive management of the bay's seagrass ecosystem.

The use of spatial linear models as the primary analytical tool is a useful tool from both statistical and ecological perspectives. Seagrass cover observations collected across a bay are inherently spatially structured, as nearby sites share similar environmental conditions, disturbance histories, and ecological communities. Models that ignore this dependence will likely provide insufficient results. By explicitly modeling spatial dependence via a spatial covariance function, the analyses presented here produce more reliable estimates of covariate effects and more honest characterizations of uncertainty than a conventional non-spatial model would provide. These models are typically also more interpretable and characterize uncertainty more effectively than machine learning models. The inclusion of EHMP subzone as a random effect further accounts for structured variation among management-relevant regions of the bay, ensuring that covariate estimates reflect within-subzone patterns rather than being confounded by between-subzone differences in baseline cover. The block Kriging helps understand bay-wide trends and can immediately characterize bay-wide predictions on the total seagrass cover scale (and not a transformed scale) – a useful and flexible framework for scenario modeling under future conditions.

The two-stage modeling framework – first modeling total cover as a function of water quality and geographic drivers, then modeling the residuals from that model as a function of site-level resilience variables – is a structured approach to separating bay-wide environmental patterns from site-specific ecological resilience capacity. This builds directly on the framework proposed by [Gilby et al., 2023] to help understand drivers of average resilience and provide useful insight needed to inform broader conservation strategies. The robustness of the findings are further support by the consistency of results across multiple modeling approaches (e.g., spatial beta regression model, random forest), which provides evidence that the relationships identified are genuine rather than resulting solely from specific modeling choices.

The results of this report provide a quantitative, spatially structured foundation for adaptive management of Moreton Bay seagrass. The understanding of important water quality and geographic variables, average predictions via block Kriging, and the study of site-level resilience indicators provides evidence that can inform targeted, context-specific intervention through the bay.

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Appendix

This is an appendix